

# Vatsal Trivedi

Principal Engineer · Generative AI Platforms · Enterprise Integrations

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## Profile

I'm a principal engineer who builds **generative AI (GenAI)** products and the platforms they run on. At Phamax, I lead Ariya's architecture and delivery across **eight pharma and life-sciences client deployments**. The platform handles source-backed generation, regulated content workflows, and enterprise integrations. Before that, I led engineering for Factoreal's omnichannel marketing product and Comviva's MobiLytix Rewards, working on analytics at scale, loyalty targeting, and event-driven services. I write full-stack code, set engineering standards, and lead teams through production releases.

## Focus areas

Generative AI platforms, Large Language Model pipelines, Retrieval-Augmented Generation, MLR (medical, legal, regulatory) review, and enterprise integrations across pharma client deployments.

## Experience

Phamax

Principal Engineer (Ariya Platform)

Bengaluru, India  
May 2025 - Present

- Ariya: generative AI platform for pharmaceutical content, research, analytics, and engagement
- I lead architecture and delivery for Ariya across eight pharma and life-sciences client deployments.
- Led system design for splitting Ariya into core API, portal, and authoring services on Azure App Service and Azure Functions, and standardized CI/CD across staging, UAT, and production for eight client deployments.
  - Architected a three-stage **Large Language Model (LLM)** generation pipeline (Content Compose) on **Azure OpenAI** with hybrid Azure AI Search retrieval, post-generation citation rebuilding, and a **14-check validator**.
  - Owned configuration-driven authoring, localization, and **MLR (medical, legal, regulatory) review**, with compliance checks and distribution across content types and markets.
  - Designed LLM orchestration with prompts, context injection, embeddings, and tool calling, and integrated Veeva CRM and Salesforce Marketing Cloud into each tenant's workflows.
  - Shipped a white-label **Retrieval-Augmented Generation (RAG)** assistant over **551 medical publications** on Azure Container Apps, with role-based access, source-PDF citations, audit logs, and an admin console.
  - Owned a streaming voice assistant (Speech Avatar) on Azure Speech and Django, including credential exchange, conversation logs, and post-call analysis.
  - Designed medallion (Bronze/Silver/Gold) ingestion with a DBML registry, run tracking, and deterministic keys for repeatable UPSERTs. Built a data catalog and **Healthcare Professional (HCP)** email-data APIs for the daily sales brief.
  - Drove **20 security fixes** across **three client release lines**, including encrypted phone numbers at rest, JWT validation, hashing, SQL access, SonarQube checks, security automation on pipeline runs, dependency audits, and database certificate rotation across seven environments and four services.

Comviva

Technical Lead

Bengaluru, India  
October 2023 - March 2025

- MobiLytix Rewards: multi-tenant loyalty and campaign platform for telecom and retail
- Technical lead for the cloud-agnostic SaaS rewrite of MobiLytix Rewards, a multi-tenant loyalty platform for telecom and retail.
- Led the multi-tenant platform rewrite in TypeScript, React, FeathersJS, PostgreSQL, Delta Lake, and vector databases, with Azure Key Vault for tenant secrets and the product features and shared services.
  - Owned Kafka campaign processing for high-throughput reward and loyalty workflows, with asynchronous execution, caching, and AI/ML personalization and segmentation.
  - Architected a JSON-to-PostgreSQL segmentation engine with **26 operators**, including 11 date operators, custom member attributes without schema changes, scheduled runs, and run history.
  - Drove transactional club provisioning so partial tenant setup rolls back, and built shared PostgreSQL and Redis clients used by five services.

- Introduced observability and Jest coverage gates, including **48 segmentation tests** and platform test coverage **above 90%**, and coordinated releases and security fixes across six services.

Factoreal

Technical Lead

Bengaluru, India

December 2021 - September 2023

Omnichannel marketing automation for ecommerce, sports, and other businesses

Led engineering for email, SMS, WhatsApp, push, social, customer journeys, commerce, and analytics.

- Owned **Google Cloud** BigQuery analytics for a platform handling **10 billion transactions**, including automated provisioning and scoring across **71 customer datasets and 998 scheduled queries**.
- Designed an audience scoring model (Fan Maturity Model) for sports: six channel scores, customer-configurable weights, dynamic tiers, and daily scoring that improved retention.
- Rebuilt the Shopify app for its App Store relaunch, including installation, signup, store provisioning, OAuth, webhook coverage from 17 to 40 topics, and GraphQL billing subscriptions.
- Built SeatGeek ticketing synchronization, WhatsApp commerce capture, and shared contact operations across five services, with country-aware phone normalization to stop duplicate contacts.
- Led the platform UI/UX revamp on Strapi, and owned the WordPress side of the product and website.

Factoreal

Senior Engineer

Bengaluru, India

June 2020 - November 2021

Omnichannel marketing automation for ecommerce, sports, and other businesses

Built data pipelines, integrations, and channel support for the core API, React portal, and backend services.

- Built PostgreSQL and Cassandra-to-BigQuery synchronization on **Google Cloud** with **Apache NiFi**, across platform microservice workflows, with incremental processing, pagination, and automated dataset provisioning at signup.
- Built email delivery and engagement KPI APIs, stored email and content in Azure Blob Storage, and embedded GoodData dashboards for deliverability, audience response, and unsubscribe rates.
- Made Shopify order ingestion safe under retries with application idempotency and database uniqueness constraints, published Razorpay payment events into Kafka, and migrated contacts from HubSpot.

Tata Consultancy Services (TCS)

Assistant System Engineer

Gandhinagar, India

September 2018 - May 2020

Life sciences work for global clients.

Built full-stack applications and analytics for global life-sciences clients in a regulated environment.

- Delivered more than 10 solutions from the start through customer deployment.
- Built React and Angular applications with Node.js and Express backends for enterprise life-sciences workflows.
- Set up Google Analytics 360, Google Tag Manager, and Google Data Studio (now Looker Studio) so life-sciences teams can measure digital activity and performance.

## Projects

openpreflight

openpreflight.xyz

Self-hosted GitHub Check Runs for private pull requests. One Go binary, one SQLite file, on a server you run. Apache-2.0.

- GitHub App, webhooks, runner, UI, and local state in one deployable unit. Docker Compose or a Linux release binary.
- The project checks itself: docs, site, and the binary each get a Check Run from ci.openpreflight.xyz. GitHub Actions is only used to publish release images.
- Wrote a five-part series on vatsal.xyz, from the trigger model through running untrusted code.

Go, SQLite, Docker, GitHub Apps

Knock

knock.codes

Copy-paste React and Tailwind components for private previews: access screens, invitations, and draft controls. Drops into a shadcn project.

React, Tailwind, shadcn/ui

InboxCraft

inboxcraft.app

A browser-only PowerShell generator for Microsoft 365 Outlook inbox rules. Quick and Advanced modes, with templates for GitHub, Slack, and Jira. Scripts stay in the browser.

TypeScript, React, PowerShell

audit-plan-tasks

github.com/trivedi-vatsal/audit-plan-tasks

Agent Skill that scaffolds AUDIT.md, PLAN.md, and TASKS.md so a plan is based on what you actually checked, not on the request.

- Audit first: read the code, run the commands, hit the real endpoints, then plan.
- Install with npx skills add trivedi-vatsal/audit-plan-tasks.

Agent Skills, Planning

Other experiments and smaller tools are at [vatsal.xyz/experiments](https://vatsal.xyz/experiments).

Skills

**Core Areas:** Generative AI, AI Platforms, Enterprise Integrations, Multi-Tenant SaaS, LLM Systems, Distributed Systems, Event-Driven Architecture, Developer Tooling, Data Engineering, Application Security, Regulated Environments, Platform Architecture, Engineering Leadership, Regulated Content Workflows

**AI & LLM Engineering:** Generative AI, Large Language Models, Retrieval-Augmented Generation, LLM Orchestration, RAG Pipelines, Prompt Engineering, Tool Calling, AI Evaluations, Azure OpenAI, Azure AI Search, Azure Speech, Vector Databases, Agent Skills

**Platform Engineering:** CI/CD, GitHub Actions, Docker, Azure App Service, Azure Functions, Azure Blob Storage, Azure Key Vault, SonarQube, Observability, Security Automation, Azure Container Apps

**Programming Languages:** TypeScript, JavaScript, Python, Go, SQL, PowerShell

**Frameworks & Tools:** React, Node.js, Django, Strapi, FeathersJS, Express, Kafka, Jest, GraphQL, REST, Shopify, WordPress

**Databases:** PostgreSQL, SQLite, BigQuery, Redis, Cassandra, Delta Lake

**Data Engineering:** Google Cloud, Apache NiFi, Medallion Architecture, ETL & Sync Pipelines, GoodData, Data Catalogs, Metadata Discovery

**Enterprise Integrations:** Veeva CRM, Salesforce Marketing Cloud, Shopify, SeatGeek, HubSpot, Razorpay

Education

**Bachelor of Engineering (B.E.), Computer Engineering**

2014 - 2018

Birla Vishvakarma Mahavidyalaya Engineering College, Vallabh Vidyanagar

Volunteer

**Creative Designer - Google Developer Groups Ahmedabad**

2018 - Present

Design for the GDG Ahmedabad chapter: event visuals, branding, and social graphics.

**Summer of Code 2020 Mentor - GirlScript**

2020 - 2020

Mentored contributors on project structure, code review, and how to land a change.

Languages

English - Proficient, Gujarati - Native, Hindi - Fluent